Project -1

CMPn361

Modern Academy

الاكاديمية الحديثة للهندسة والتكنولوجيا

Real Estate Company (Network)

DR: Seham Muawad Ali

DR: Musheera

Student:

- Abdelrahman Mahmoud Ali Ahmed
- Yousef Adel Emarra Hassan
- Abdelrahman Yasser Mohamed ELbasuoni
- Saif Alden Qotp Said Qotp
- Soheil Yousry Abdelwahed Ibrahim
- Mohamed Ahmed Abdelaleem Mostafa
- Ali Ahmed Mohamed

Table of Contents

Summary:

This system simulates the internal and external network systems of the company to improve the quality of production and good tracking of performance Company personnel and production tracking.

There is a human resources and developers Department in two main branches, Cairo branch and Giza branch.

Additions

Developing internal and external servers and networks and working on strong, fast and streamlined performance

Technical Plan:

1-Hardware:

- Networking Hardware

  - Routers: Direct data between networks.

  - Switches: Connect devices within a single network.

  - Modems: Convert digital data to analog for transmission over phone lines.

2-Software:

We just use whatever software the academy must reduce the cost and increase the revenue. Some of the software is even ready with computers. The software that we used in the development process is:

- Windows 10/11
- Cisco Packet Tracer
- TCP/IP: Core protocol suite for the internet.
- HTTP/HTTPS: Protocols for web browsing.
- FTP: Protocol for file transfers.

Management Plan

project Schedule

- The unexpected always happens. Always contingency in planning. Allocations of tasks are divided into a few phases, which include system analysis, design, development, and implementation. Each member in the team has their respective task to complete in the required time frame. Below is the position of the team personnel:

- Project Leader:
- *Co-ordination of work in general
- *Scheduling of project activities
- *Motive members
- *Delegation of duties

IT Programmers:

*Scheduling programming jobs

*Perform research on technical details for system development

*Ensure program function meet users need

Feasibility study:

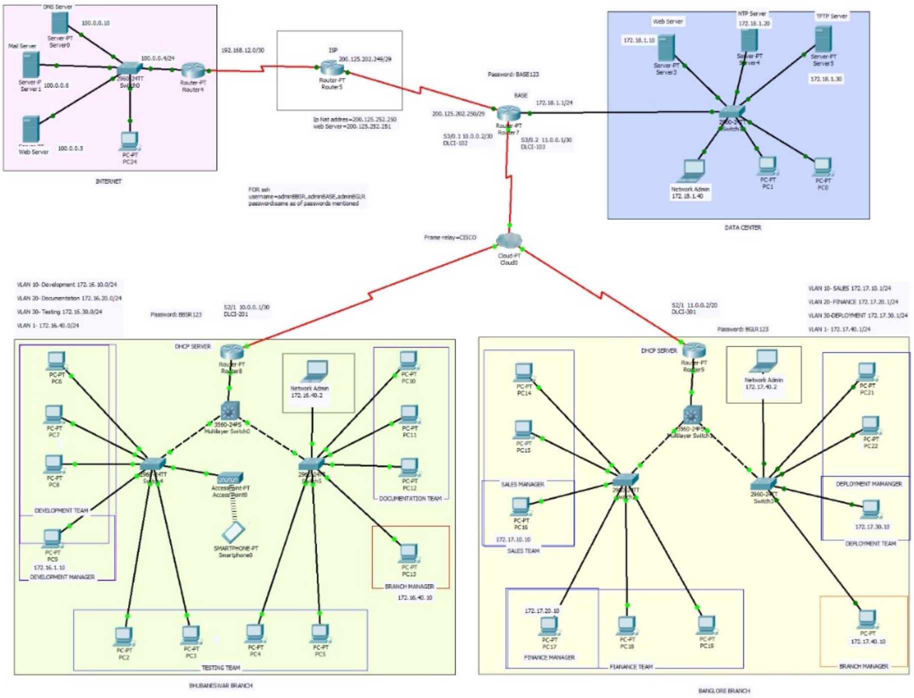
Pricing:

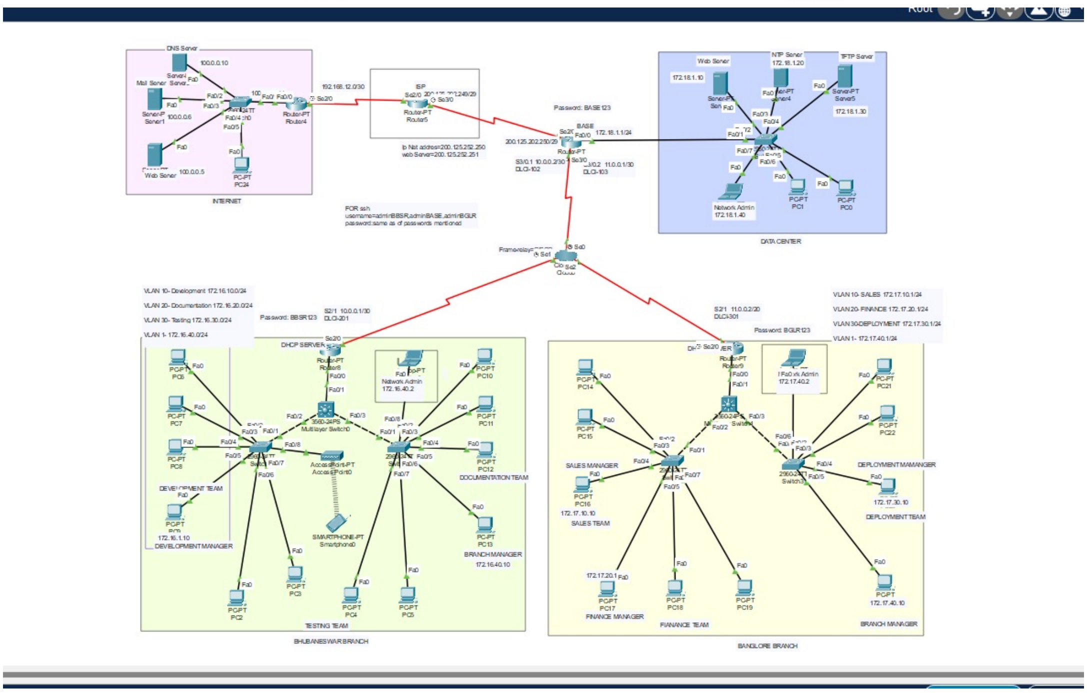
Sub Total
$11,000.00

Product Name	Quantity	Price	Unit	Total
server	1.00	$ 3,000.00	0.00	$ 3,000.00
Coding and Testing	1.00	$ 6,500.00	0.00	$ 6,500.00
Hosting, Monitoring, and Updating	1.00	$ 1,500.00	0.00	$ 1,500.00

TOTAL USD 11,000.00

Design :


INTERNET
ISP
BASE
DATA CENTER
BHUBANESWAR BRANCH
BANGLORE BRANCH


INTERNET
DATA CENTER
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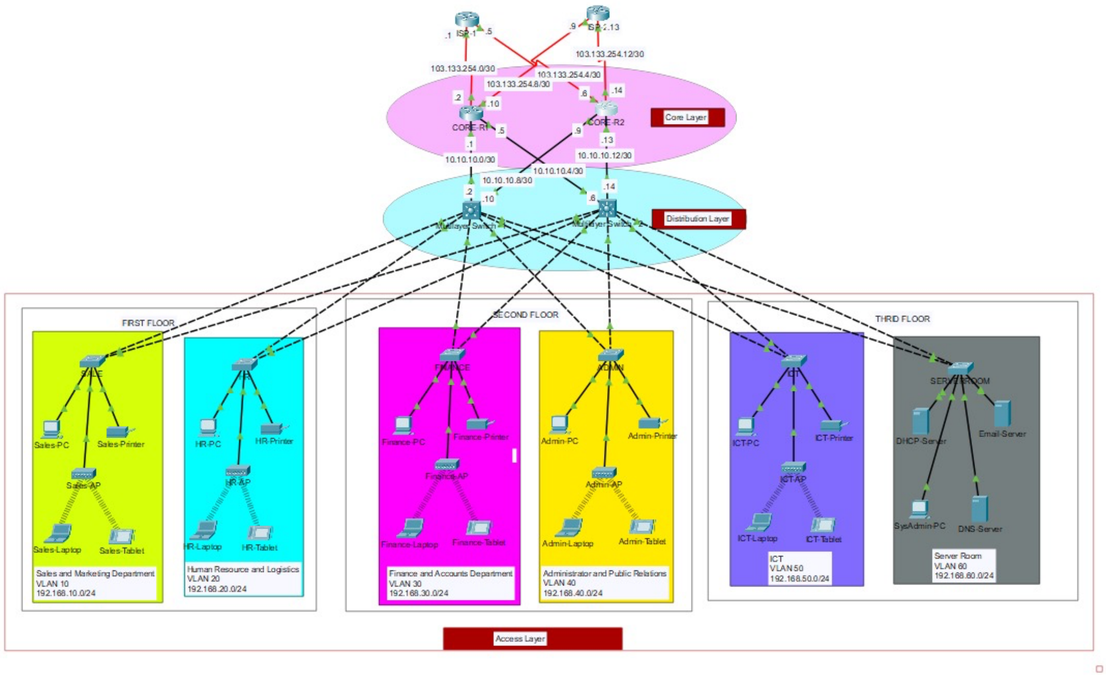
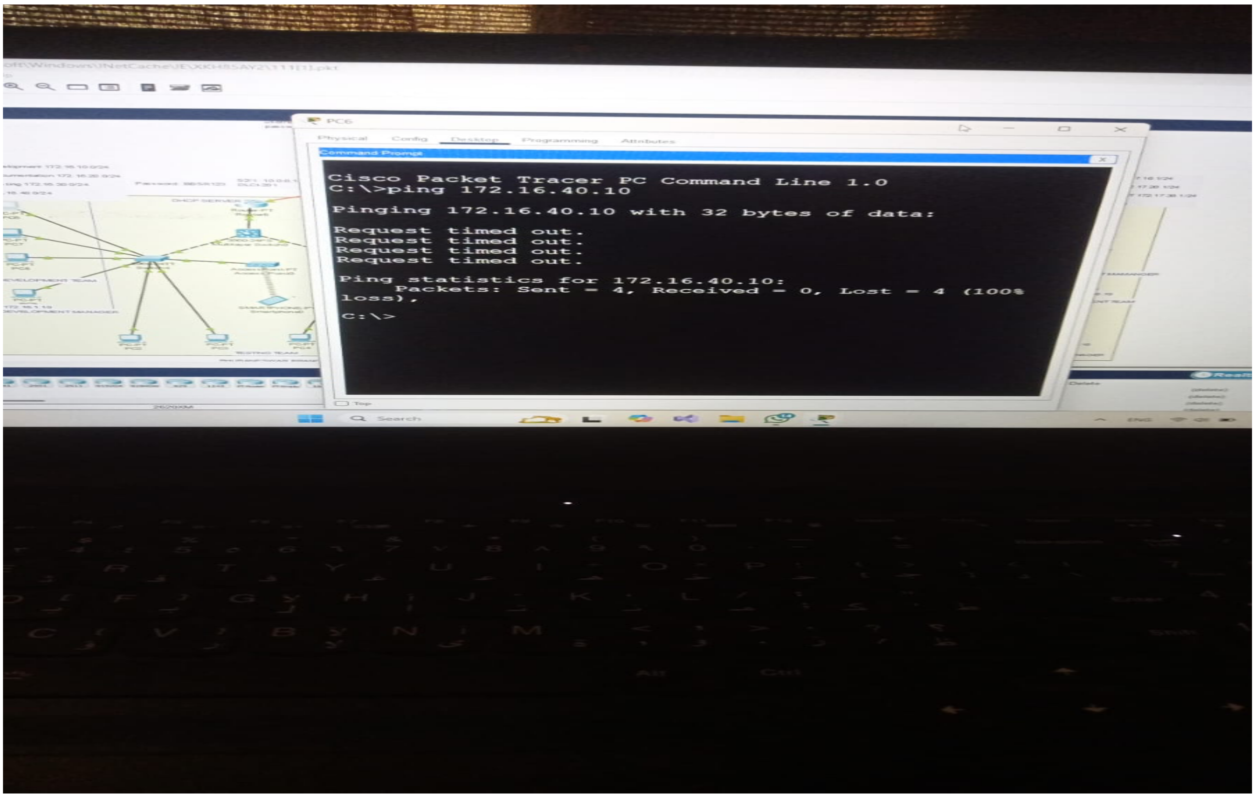
BANGLORE BRANCH



ISP-1
ISP-2
103.133.254.0/30
103.133.254.4/30
103.133.254.8/30
103.133.254.12/30
CORE-R1
CORE-R2
Core Layer
10.10.10.0/30
10.10.10.4/30
10.10.10.8/30
10.10.10.12/30
Distribution Layer
FIRST FLOOR
SECOND FLOOR
THIRD FLOOR
SALES
FINANCE
ADMIN
ICT
SERVER ROOM
Sales-PC
Sales-Printer
Sales-AP
Sales-Laptop
Sales-Tablet
HR-PC
HR-Printer
HR-AP
HR-Laptop
HR-Tablet
Finance-PC
Finance-Printer
Finance-AP
Finance-Laptop
Finance-Tablet
Admin-PC
Admin-Printer
Admin-AP
Admin-Laptop
Admin-Tablet
ICT-PC
ICT-Printer
ICT-AP
ICT-Laptop
ICT-Tablet
DHCP-Server
Email-Server
SysAdmin-PC
DNS-Server
Sales and Marketing Department
VLAN 10
192.168.10.0/24
Human Resource and Logistics
VLAN 20
192.168.20.0/24
Finance and Accounts Department
VLAN 30
192.168.30.0/24
Administrator and Public Relations
VLAN 40
192.168.40.0/24
ICT
VLAN 50
192.168.50.0/24
Server Room
VLAN 60
192.168.60.0/24
Access Layer



PC6
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.40.10
Pinging 172.16.40.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 172.16.40.10:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>

Components:

The network design for the project incorporates the following devices:

Routers :

- 2 ISP router for upstream connectivity.
- Positioned at the core layer for redundancy.
- Connect to both ISPs for internet connectivity.
- Configured with static, public IP addresses from ISPs.

Multilayer Switches :

- Deployed at the core layer to provide redundancy and efficient routing.
- Configured for both switching and routing functionalities.
- Assigned IP addresses to enable inter-VLAN route

Distribution Layer Switches (Multiple):

- Connect individual departments to the core layer.
- Facilitate communication within respective VLANs.

2. **End-User Devices (PCs):**

- Deployed at the access layer.
- Connected to distribution layer switches for departmental access.

3. **Cisco Access Points (APs):**

- Positioned at the access layer to provide wireless connectivity.
- Ensure wireless network availability in each department.

4. **DHCP Servers (1):**

- Located in the server room.
- Dynamically allocate IP addresses to end-user devices.

5. **Server Room Devices (Servers, etc.):**

- DNS server, HTTP server etc.
- Devices in the server room are allocated static IP addresses.

These devices may include servers, storage units, and networking equipment

Attacks:

1. Man-in-the-Middle Attack (MITM)

Attack Definition:
In this attack, the attacker intercepts or manipulates communications between two or more devices, allowing them to eavesdrop on or alter the data being sent between the parties.

How to Mitigate the Attack:

- Use Encryption:
  Encryption protects the data while it is in transit, making it difficult for the attacker to read or alter the information.
- Enable HTTPS:
  Ensure that HTTPS or TLS is used for secure communication between devices.
- Use a VPN:
  A Virtual Private Network (VPN) provides an additional layer of security for internet communications.

Commands for Protection (in case of VPN setup):

- On a Cisco Router:

bash

Copy code

```
crypto isakmp policy 10

encryption aes 256

authentication pre-share
```

---

2. Denial of Service (DoS) / Distributed Denial of Service (DDoS) Attack

Attack Definition:
DoS or DDoS attacks flood a server or network with an overwhelming amount of traffic, causing the system to become unavailable to legitimate users.

How to Mitigate the Attack:

- Enable Traffic Filtering:
  Use tools like ACLs (Access Control Lists) to filter suspicious traffic and block malicious IP addresses.
- Implement Load Balancing:
  Use load balancing to distribute traffic evenly across multiple servers, preventing overload on a single server.
- Rate Limiting:
  Configure rate limiting to restrict the number of requests a user can make to the server in a given time period.

Example Commands for Mitigation (on Cisco Router):

- To configure ACL to block malicious IPs:

bash

Copy code

```
access-list 101 deny ip 192.168.1.100 0.0.0.255 any

access-list 101 permit ip any any

interface gigabitEthernet 0/1

ip access-group 101 in
```

---

3. IP Spoofing Attack

Attack Definition:
The attacker falsifies the source IP address of packets to appear as though they are coming from a trusted source, allowing unauthorized access to systems or services.

How to Mitigate the Attack:

- Use Ingress and Egress Filtering:
  Apply ACLs to filter incoming and outgoing traffic based on IP

addresses to prevent spoofed packets from entering or leaving the network.

- Implement Source Address Validation:
  Ensure that all incoming packets have valid source IP addresses.

Example Commands for Mitigation:

- To filter IP spoofing using an ACL:

bash

Copy code

```
access-list 100 deny ip 192.168.0.0 0.0.255.255 any

access-list 100 permit ip any any

interface gigabitEthernet 0/1

ip access-group 100 in
```

---

4. ARP Spoofing Attack

Attack Definition:
In this attack, the attacker sends fake ARP (Address Resolution Protocol) messages over a local network, associating their MAC address with the IP address of another device, allowing them to intercept traffic meant for that device.

How to Mitigate the Attack:

- Static ARP Entries:
  Set static ARP entries on devices to prevent attackers from modifying the ARP table.
- Use Dynamic ARP Inspection (DAI):
  Enable DAI on the switch to verify ARP packets and prevent spoofing attacks.

Example Commands for Mitigation (on Cisco Switch):

- To configure static ARP entries:

bash

Copy code

arp 192.168.1.1 00-1D-A1-00-01-01

- To enable Dynamic ARP Inspection (DAI):

bash

Copy code

ip arp inspection vlan 10

---

5. Port Scanning Attack

Attack Definition:
Port scanning is a technique used by attackers to discover open ports on devices in a network, which may reveal vulnerabilities that can be exploited.

How to Mitigate the Attack:

- Use Port Security:
  Configure Port Security on switches to restrict which devices can connect to network ports and prevent unauthorized access.
- Disable Unused Ports:
  Disable any unused network ports on switches to reduce the attack surface.
- Implement Firewalls and Intrusion Detection Systems (IDS):
  Use firewalls and IDS to monitor and block suspicious port scanning activities.

Example Commands for Mitigation:

- To enable Port Security on a specific port (e.g., FastEthernet 0/1):

bash

Copy code

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport port-security
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security violation restrict
```

- To disable unused ports:

bash

Copy code

```
Switch(config)# interface range fa0/10 - 24
Switch(config-if-range)# shutdown
```

---

6. MAC Flooding Attack

Attack Definition:
In this attack, the attacker sends a large number of fake MAC addresses to a switch, causing it to overflow its MAC address table and begin broadcasting traffic to all ports, which can lead to a loss of network performance or interception of sensitive data.

How to Mitigate the Attack:

- Enable Port Security:
  Use Port Security to limit the number of MAC addresses that can be learned on a switch port.
- Implement Dynamic MAC Address Learning Limits:
  Set limits on the number of MAC addresses a port can learn.

Example Commands for Mitigation:

- To configure Port Security to limit MAC addresses:

bash

Copy code

```
Switch(config)# interface fa0/1
Switch(config-if)# switchport port-security maximum 2
Switch(config-if)# switchport port-security violation shutdown
```

---

7. DNS Spoofing Attack

Attack Definition:
DNS Spoofing (or DNS Cache Poisoning) involves an attacker corrupting the DNS cache to redirect traffic to malicious websites.

How to Mitigate the Attack:

- Use DNSSEC (DNS Security Extensions):
  Implement DNSSEC to digitally sign DNS records and prevent tampering.
- Configure DNS Filtering:
  Use DNS filtering solutions to block access to known malicious domains.
- Configure Strict DNS Caching Policies:
  Reduce the cache time for DNS entries to limit the impact of poisoned records.

Commands for Mitigation (on Cisco Router using DNS filtering):

- To configure DNS filtering:

bash

Copy code

```
ip dns server
ip dns domain-lookup
ip name-server 8.8.8.8
```

Commend:

1. Subnetting and VLSM (Variable Length Subnet Masking)

VLSM allows for more efficient IP address utilization by using different subnet masks within the same network.

Example:

- Given IP: 192.168.1.0/24, you want to create two subnets:
  - Subnet 1: 192.168.1.0/25
  - Subnet 2: 192.168.1.128/25

Commands (on Cisco Router):

```
ip subnet-zero      # Enable subnet-zero (if needed)
ip address 192.168.1.1 255.255.255.128   # For Subnet 1
ip address 192.168.1.129 255.255.255.128 # For Subnet 2
```

---

2. VLAN Configuration

VLANs are used to segment networks into different broadcast domains.

Commands (on Cisco Switch):

vlan 10

name Finance

exit

vlan 20

name HR

exit

interface range fa0/1 - 24

switchport mode access

switchport access vlan 10

---

3. EtherChannel Configuration

EtherChannel aggregates multiple physical links between two switches into a single logical link for increased bandwidth and redundancy.

Commands (on Cisco Switch):

interface range gi0/1 - 2

channel-group 1 mode active   # Use 'active' for LACP (Link Aggregation Control Protocol)

---

4. Spanning Tree Protocol (STP) Configuration

STP is used to prevent loops in the network by managing the forwarding paths of switches.

Commands (on Cisco Switch):

spanning-tree vlan 1 priority 4096 # Set lower priority to make this switch root

spanning-tree portfast # Enable PortFast on access ports

spanning-tree bpduguard enable # Enable BPDU Guard on access ports

---

5. PortFast Configuration

PortFast is used to allow a port to transition from blocking to forwarding immediately, bypassing the STP process.

Commands (on Cisco Switch):

interface fa0/1

spanning-tree portfast

---

6. OSPF (Open Shortest Path First) Configuration

OSPF is a dynamic routing protocol used for distributing routing information within an autonomous system.

Commands (on Cisco Router):

router ospf 1

network 192.168.1.0 0.0.0.255 area 0 # Define OSPF network and area

---

7. EIGRP (Enhanced Interior Gateway Routing Protocol) Configuration

EIGRP is a Cisco proprietary dynamic routing protocol that uses a distance vector approach.

Commands (on Cisco Router):

router eigrp 1

network 192.168.1.0 0.0.0.255   # Define EIGRP network

no auto-summary   # Disable automatic summarization

---

8. Port Security Configuration

Port Security limits the number of MAC addresses allowed on a switch port to enhance security.

Commands (on Cisco Switch):

interface fa0/1

switchport port-security

switchport port-security maximum 2   # Allow only 2 MAC addresses

switchport port-security violation restrict # Action on violation

---

9. ASA Firewall Configuration

The Cisco ASA (Adaptive Security Appliance) firewall provides security through access control and inspection of traffic.

Commands (on ASA Firewall):

access-list outside_access_in extended permit ip any any

access-group outside_access_in in interface outside

---

10. VTP (VLAN Trunking Protocol) Configuration

VTP allows switches to share VLAN information across the network.

Commands (on Cisco Switch):

vtp mode server   # Set switch as VTP server (default mode)

vtp domain example.com # Set the VTP domain name vtp password mypassword # Set the VTP password

---

11. VPN Configuration (IPSec VPN)

VPNs provide secure communication over the internet by encrypting data between devices.

Commands (on Cisco Router for IPSec VPN):

```
crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 2
key mysharedkey

crypto ipsec transform-set myset esp-aes-256 esp-sha-hmac

crypto map mymap 10 ipsec-isakmp
set peer 192.168.2.1
set transform-set myset
match address 101

interface gigabitEthernet 0/0
crypto map mymap
```

---

12. BPDU Guard Configuration

BPDU Guard protects a network from potential loops by shutting down ports that receive BPDUs (Bridge Protocol Data Units).

Commands (on Cisco Switch):

interface range fa0/1 - 24

spanning-tree bpduguard enable

---

13. BPDU Filtering

BPDU Filtering prevents a port from sending or receiving BPDU frames, which is used to prevent unnecessary participation in STP.

Commands (on Cisco Switch):

interface range fa0/1 - 24

spanning-tree bpdufilter enable

Summary of Key Commands:

- Subnetting & VLSM: ip address x.x.x.x x.x.x.x
- VLAN: vlan [VLAN_ID], switchport mode access, switchport access vlan [VLAN_ID]
- EtherChannel: channel-group [Group_Num] mode [active|passive]
- STP: spanning-tree vlan [VLAN_ID] priority [Value], spanning-tree portfast
- PortFast: spanning-tree portfast
- OSPF: router ospf [Process_ID], network [IP] [Wildcard_Mask] area [Area_ID]
- EIGRP: router eigrp [AS_Num], network [IP] [Wildcard_Mask]

- Port Security: switchport port-security, switchport port-security maximum [Num]
- ASA Firewall: access-list [Name] extended permit ip any any, access-group [Name] in interface [Interface]
- VTP: vtp mode [server|client|transparent], vtp domain [domain_name], vtp password [password]
- VPN: crypto isakmp policy, crypto ipsec transform-set
- BPDU Guard: spanning-tree bpduguard enable
- BPDU Filtering: spanning-tree bpdufilter enable

These commands are essential for configuring and securing various network services like routing, switching, security, and VPNs.

Future vision

- Network Monitoring Tools
- Enhanced Security Measures
- Virtualization Technologies
- Advanced Routing Protocols
- IPv6 Implementation
- Wireless Network Expansion
- Cloud Integration
- Ongoing Training and Skill Development
- Regular Security Audits